STONE: Soft-Label Tone Classification for Taiwanese Hokkien via CTC-Guided Contrastive Prototypes on Self-Supervised Models

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Abstract

The complex tone and tone sandhi system of Taiwanese Hokkien presents significant challenges to TTS training and evaluation, and there is a lack of large-scale research on the classification and evaluation of the key tone attribute of Taiwanese Hokkien based on neural networks. We propose a Soft-label TONE classification for Taiwanese Hokkien via a CTC-guided prototype learning (STONE) architecture. This architecture combines an SSL pre-trained feature extractor with a dual-stream downstream model, comparing tone error rate and alignment accuracy between using only the output layer, adaptive weighting of intermediate layers, and fine-tuning or freezing the pre-trained model. Fisher Criterion and Linear Probing Accuracy are used to analyze the tone information contained in different layers and the downstream projector output. Experimental results demonstrate that all backbones achieve an in-distribution Tone Error Rate (TER) below 3% after fine-tuning, and that the fine-tuned XLSR-Chinese backbone yields the best overall performance, with a 22.11% out-of-distribution (OOD) Top-1 TER, a 5.76% OOD Top-3 TER, and a 74.68% alignment accuracy. By generating soft-label tones to accommodate natural speech ambiguities, STONE provides quantifiable confidence scores, laying a critical foundation for rigorous future evaluation benchmarks.

Index Terms: tone classification, Taiwanese Hokkien, self-supervised speech representation

1. Introduction

Taiwanese Hokkien is a typical low-resource tonal language used by millions of people. Compared to other tonal languages, Taiwanese Hokkien has extremely complex phonological characteristics, including seven tones and a highly dynamic tone sandhi. In natural speech, the tone of each syllable in the original text needs to undergo complex changes according to grammatical structure and semantics. Therefore, accurate automatic tone classification of Taiwanese Hokkien is not only key to evaluating Taiwanese Hokkien TTS, but also a major challenge in speech representation learning.[1]

A core challenge in tone classification lies in tone sandhi. Because tone sandhi results are not unique depending on the speaker's pauses within long sentences, there is no absolute ground truth tone sandhi result for a given text. Furthermore, the dataset only contains the original (un-sandhied) tones, which means there is a mismatch between the labeled tone and the actual surface tone of each syllable. To address this, we utilized an existing Taiwanese Hokkien words speech dataset. By directly applying tone sandhi rules to texts containing only a few syllables, we were able to obtain relatively reliable tone labels for training.

Also, due to limited corpus resources, the forced-alignment results from the Montreal Forced Aligner (MFA)[2] are not accurate enough for tone boundaries. Since tone is essentially a change in pitch and duration, inaccurate segment boundaries will fundamentally affect tone classification. Therefore, we first use CTC[3, 4] to directly align each syllable of the input audio.

While using SSL pre-trained models in speech processing tasks is a common practice in related research[5], there is a lack of research on the ability of common pre-trained models to process Taiwanese Hokkien tone information. Therefore, we conduct experiments to reveal the shortcomings of the native tone representations of self-supervised models, and to demonstrate the ability of different SSL pre-trained feature extractors to process Taiwanese Hokkien tones after fine-tuning.

In natural speech, there may be reasonable tone inaccuracies. However, hard-label tone cannot measure the closeness of a syllable to each tone, and therefore cannot discuss whether tone errors are reasonable biases or serious errors. Therefore, we use the alignment results to train the projector and perform supervised contrastive learning through triplet loss[6, 7, 8] to obtain a prototype[9] that can generate soft labels.

2. Background: Tonal System

2.1. Tonal Inventory

Taiwanese Hokkien has evolved directly from Old Chinese and is influenced by Middle Chinese, which has an eight-tone system. These tones are numbered 1 to 8, but in Taiwanese Hokkien, tone 6 has been merged into tone 7. In addition, Taiwanese Hokkien has a neutral tone. Although it is generally not assigned a specific tone number, it is marked as zero in this study. Furthermore, when three identical single Chinese characters are linked together for emphasis or exaggeration, the first character becomes the significantly higher ninth tone. Although not uncommon in daily conversation, this situation is very rare in our data, so this case is omitted in this study [10, 11, 12].

2.2. Tone Sandhi Rules and Dialectal Variations

In Taiwanese Hokkien, syllables except those preceding open juncture require tone sandhi. However, open juncture can appear in many places within a sentence, with its occurrence depending on the speaker's intended meaning and pauses. Like other Sinitic languages, Taiwanese Hokkien often combines multiple syllables to form a word that conveys a complete meaning; therefore, open juncture will not appear in the middle of such words, and these words are connected by hyphens in the phonetic transcription. Furthermore, syllables before a neutral tone do not require tone sandhi [13].

Therefore, it can be summarized that if a syllable is followed by a hyphen, tone sandhi is mandatory; if a syllable is followed by a punctuation mark or two hyphens, tone sandhi is prohibited; in other cases, where a syllable is followed by a blank, tone sandhi may or may not occur. However, in prin-

ciple, except for a few valid open junctures for breaks, tone sandhi should occur as long as the speech is continuous. The actual changes during tone sandhi are shown in Table 1.

Table 1: *Tone sandhi rules for Taiwanese Hokkien.[12, 14]*

Base	**1**	**2**	**3**	**4**	**5**	**7**	**8**	**0**
Zhangzhou	7	1	2	8, 2	7	3	4, 3	0
Quanzhou	7	1 or 5	2	8, 2	3	3	4, 3	0

The tone sandhi result for tone 2 in Quanzhou dialect varies depending on the region; tone 1 and tone 5 are both present. For entering tones (4 and 8), the sandhi target differs by final consonant: *-p/-t/-k* vs. glottal stop *-h*.

3. Proposed Tone Classification Engine

In our tone classifier, we employ an SSL pre-trained model as a backbone. The features extracted by this model are processed using either the direct output layer or a weighted sum of selected layers with learnable weights [5]. The features are first fed into an LM head to obtain an output sequence over seven tones, a neutral tone, and a blank token, which is used for greedy decoding to align each syllable. Mean pooling is then performed over the grouped SSL features, and the pooled vectors are projected to obtain the final tone prototypes. The overall architecture is shown in Fig. 1.

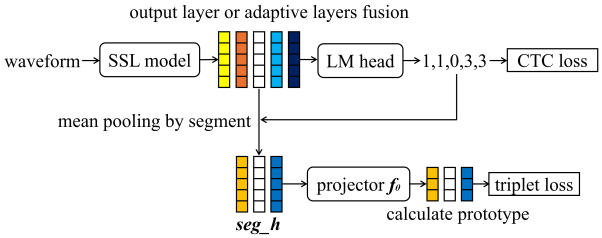


Figure 1: *Model architecture.*

3.1. Training

We train with audio and its true tone labels on the dictionary dataset, which provides more accurate tone sandhi results (see the Datasets section). The LM head output is used to compute the CTC loss, while its greedy decoding gives the segment boundaries that mean-pool the SSL features into the segment-vector sequence seg_h, whose i-th element seg_h_i denotes one segment.

Then, inputting this result into the projector f_θ yields an L2-normalized representation on a 128-dimensional unit hypersphere: $e_i = f_\theta(seg_h_i)/\|f_\theta(seg_h_i)\|_2$.[6] On this hypersphere, we use Euclidean distance $D(e_i, e_j) = \|e_i - e_j\|_2 = \sqrt{2 - 2\cos(e_i, e_j)}$ to measure the features.

During this training phase, starting from the third epoch, we begin training the projector and calculating triplet loss to avoid poor alignment quality in the early stages of training. When calculating triplet loss, we first extract $K = \min(|\mathcal{S}|, |\mathcal{Y}_{ref}|)$ syllables from each sample $\mathcal{S}$, where $|\mathcal{S}|$ is the number of CTC-decoded segments and $|\mathcal{Y}_{ref}|$ the number of reference syllables, and use the tone labels y_i of the first K syllables as the correct prediction. Finally, we aggregate the feature vectors e_i of each syllable from all samples in a batch, and for each e_i, we find the farthest positive e_p for the same tone and the closest negative e_n for different tones. The triplet loss $\mathcal{L}_{triplet}$ is calculated as follows:

$$\frac{1}{M}\sum_{i=1}^{M}\max\left(0, \max_{y_p=y_i} D(e_i, e_p) - \min_{y_n\neq y_i} D(e_i, e_n) + m\right) \tag{1}$$

Here, $M = \sum_{\mathcal{S}\in\mathcal{B}} K$ is the total number of anchor syllables aggregated over all samples $\mathcal{S}$ in a batch $\mathcal{B}$, and m is a predefined triplet margin. The final loss function is given by $\mathcal{L}_{joint} = \mathcal{L}_{CTC} + \lambda \cdot \mathcal{L}_{triplet}$.

3.2. Soft Label Generation

To address the issue that hard labels yield only all-correct or all-incorrect results and cannot capture tones that are near-correct but slightly inaccurate, or that fall between two tones in natural speech, our model generates soft labels that span all possible tones.

Using the embeddings e_i defined above, we calculate the centroid of each tone over all data from the training dataset. Then, for each inference syllable, we likewise compute its embedding and its L2 distances to the eight centroids. Sorting these as $D_1 \leq D_2 \leq \cdots \leq D_8$, the tone of D_1 is the most likely, that of D_2 the second most likely, and so on.

We then define two confidence indicators. For the k-th closest tone, $score_k = D_1/D_k \in (0, 1]$ decreases as k increases and quantifies its confidence relative to the top candidate. For the k-th closest tone with $1 \leq k \leq 7$, $margin_k = (D_{k+1} - D_k)/(D_{k+1} + D_k + \epsilon)$, with ϵ a small constant for numerical stability, measures its separation from the next-ranked tone.

In this study, the top three possible tones of the soft labels are primarily used to handle syllables that are difficult to distinguish in natural speech, thus better reflecting the classification ability of the classifier. The score and margin, when used in the application and future work to establish tone accuracy benchmarks, serve as indicators to better measure the confidence level of a syllable belonging to a specific tone and the discrepancy with the next possible tone, respectively.

4. Experiment

4.1. Datasets

We used two Taiwanese Hokkien speech corpora with text labels, both providing romanization annotated with the original tone.

The first dataset was crawled from the Ministry of Education's Taiwanese Hokkien dictionary website. Therefore, most of the data in this dataset consists of Taiwanese Hokkien words with two to four syllables. Assuming there are no open junctures, directly applying tone sandhi rules to the original tone is reasonable. Therefore, this dataset can be used as a training dataset, and 15% of the data are separated for ID (in distribution) validation [15].

The second dataset, SuiSiann [16], consists of various sentences. Therefore, most of the speeches are relatively long, containing dozens of syllables, difficult to directly obtain the actual tone sandhi results. Therefore, for syllables that may or may not change tone, all valid results are considered correct. For syllables that must change tone or are prohibited from changing tone, only results conforming to the rules are allowed. Due to the uncertainty of the results, this dataset is only used for OOD (out of distribution) validation. Table 2 summarizes the statistics.

Table 2: *Dataset Statistics. (ID split: 85% Train / 15% Val)*

Dataset	Hours	Utterances	Syllables	Speakers
MoE Dict (Train)	8.207	17,859	38,889	Multi*
MoE Dict (ID Val)	1.448	3,152	6,902	Multi*
SuiSiann (OOD)	3.626	2,919	42,542	1

*Metadata lacks speaker IDs, but audio contains multiple speakers.

4.2. Metric

To measure the accuracy of tone classification, we use the TER (tone error rate) metric. However, the calculation method differs between the ID validation set and the OOD validation set. Since the ID validation set generates a reasonable and unique ground truth, the TER calculation directly uses the edit distance between the model's predicted tone sequence and the actual tone sequence, and finally performs a weighted average based on the length of the actual tone sequence. For OOD speech, multiple sandhi outcomes are permissible; thus, any rule-compliant tone is accepted as correct when calculating TER to accommodate natural ambiguity. The calculation of OOD TER only considers the case where the sequence length generated by CTC aligned decoding is the same as the true length; we denote the success rate of this length-matching alignment as OOD Align Acc, which is reported alongside OOD TER in Table 3. Also, OOD TER is divided into Top-1 calculation, which considers a correct tone as long as the classifier predicts that the nearest tone is the correct tone, and Top-3 calculation, which considers a correct tone as long as the classifier predicts that the three nearest tones have an intersection with the correct tone. Since the text length is inherently longer than the training set and aligning to near-natural speech flow is more difficult, the probability of alignment errors can be expected to be higher.

To better discuss the amount of tone information contained in each representation vector, i.e. the output after the projector, and different layers of the pre-trained model, we also used FC (Fisher Criterion), the ratio of between-class to within-class scatter across tones[17]. In addition, we used LPA (Linear Probing Accuracy) and NCA (Nearest Centroid Accuracy). For both, the whole ID dataset is split 80/20, and within each portion only the syllables that the analyzed model itself aligns with a matching length are used; LPA verifies the separation of the representation space under Multinomial Logistic Regression[18, 19], while NCA finds the center of each tone in the analyzed representation and predicts the tone by the nearest centroid.

4.3. Experimental Setup

First, we compared five pre-trained backbones: Wav2Vec2-Base[20], Wav2Vec2-Large[20], XLSR-53[21], MMS-1B-All[22], and XLSR-Chinese[23], the last of which is fine-tuned on a Mandarin Chinese speech corpus. Strategies involving either using the final output layer or a multi-layer weighted sum (we select layers 1, $N/4$, $N/2$, $3N/4$, and N, where N is the number of transformer layers and layer N is the output layer) were implemented, along with training experiments for fine-tuning and freezing the pre-trained model, totaling 20 sets of experiments. Training lasted 50 epochs with a batch size of 4, resulting in approximately 4.5k update steps per epoch. The learning rates for both the LM head and the projector were set to 10^{-3}. If the pre-trained model was fine-tuned, its learning rate was set to 2×10^{-5}. All learning rates were linearly annealed to 0 after the last epoch. The network was optimized using the AdamW optimizer [24, 25] with default betas $\beta_1 = 0.9, \beta_2 = 0.999$. The remaining hyperparameters for the triplet loss are set as $m = 0.3$, $\lambda = 0.1$.

5. Results

Currently, there is no publicly available Taiwanese Hokkien tone classification system for direct comparison. The SSL backbone comparison in this study constitutes an internal comparison, and the TER results can also serve as a reference point for future work.

5.1. Overall Performance

Table 3 summarizes the performance of different SSL backbones, layer representation modes, and feature extractor training strategies. Overall, fine-tuned XLSR-Chinese with last-layer features performs best, with a 2.22% ID TER, a 22.11% OOD Top-1 TER, a 5.76% OOD Top-3 TER, and a 74.68% alignment accuracy. This may be partly due to the multilingual pre-training it inherits from XLSR-53, which covers many tonal languages, and to the subsequent fine-tuning on Mandarin, a related Sinitic language. Indeed, XLSR-Chinese outperforms the original XLSR-53 in most settings, suggesting that the Mandarin adaptation provides a useful bias for modeling syllable-level tonal structure. However, this interpretation should be treated cautiously, as multilingual coverage, model capacity and depth are not disentangled in the current comparison.

Notably, the larger Wav2Vec2-Large underperforms Wav2Vec2-Base in terms of TER in all settings, suggesting that, with similar model architectures and pre-training data, scaling up the model does not necessarily lead to better handling of Taiwanese Hokkien speech. This observation aligns with the findings of [26].

Fine-tuning substantially reduces both ID and OOD TER across all SSL backbones and layer modes, suggesting that task-specific adaptation plays an important role in capturing syllable-level tonal information in Taiwanese Hokkien. After fine-tuning, the two layer modes differ by at most 0.4% absolute in ID TER and show no consistent direction on the OOD metrics, indicating that the choice of layer representation has limited impact once the backbone is adapted.

The OOD TER was substantially lower when using the top-3 soft labels compared to using only the highest-probability label, indicating that tone ambiguity exists in the OOD dataset, and that soft labels can indeed further identify potential tone.

5.2. Component Ablation

We compare two simplified variants, both fine-tuned from the pre-trained checkpoint with cross-entropy and replacing the projector and the prototypes with a single linear classifier over the pooled features of the same last-layer XLSR-Chinese backbone. MFA boundaries give a 12.17% ID TER, a 32.35% OOD Top-1 TER and a 7.44% OOD Top-3 TER, whereas CTC alignment, trained jointly with the CTC objective, gives 2.19%, 22.57% and 6.00% with a 72.80% OOD alignment accuracy, on par with the full system. MFA aligns 96.6% of the utterances given the reference transcript, but its OOD numbers are reported on the 74.68% subset that the full system aligns, while each CTC-based system uses its own length-matched subset. Since the variants share the backbone, the pooling scheme and the classifier, the gap comes from the inaccurate boundaries MFA produces on this corpus: boundary quality dominates accuracy and remains the bottleneck even with CTC. The projector instead reorganizes the representation space, as analyzed below.

Table 3: *Experimental Results*

Backbone Training	Layer Mode	Pre-trained Backbone	ID TER (%) ↓	OOD TER (%) ↓		OOD Align Acc (%) ↑
				Top-1	Top-3	
Frozen	Last Layer	Wav2Vec2-Base	44.19	61.01	28.22	8.77
		Wav2Vec2-Large	93.61	-	-	0.17
		XLSR-53	97.45	90.20	62.75	1.03
		XLSR-Chinese	**9.53**	35.45	13.98	34.50
		MMS-1B-All	10.29	**26.65**	**7.23**	**60.77**
	Weighted Mix	Wav2Vec2-Base	41.15	66.14	31.61	7.47
		Wav2Vec2-Large	98.77	-	-	0.00
		XLSR-53	16.81	55.66	26.19	22.64
		XLSR-Chinese	9.97	37.43	16.06	21.27
		MMS-1B-All	**7.69**	**31.65**	**11.65**	**43.17**
Fine-tuned	Last Layer	Wav2Vec2-Base	2.45	24.96	5.89	63.75
		Wav2Vec2-Large	2.56	28.84	9.18	68.28
		XLSR-53	**2.01**	23.88	6.42	72.63
		XLSR-Chinese	2.22	**22.11**	**5.76**	**74.68**
		MMS-1B-All	2.07	22.61	6.85	69.06
	Weighted Mix	Wav2Vec2-Base	2.64	27.82	7.99	64.71
		Wav2Vec2-Large	2.94	34.56	9.81	58.79
		XLSR-53	2.13	24.60	6.83	72.87
		XLSR-Chinese	2.22	22.56	**6.20**	**73.00**
		MMS-1B-All	**2.10**	**21.80**	6.35	67.73

5.3. Representation Analysis

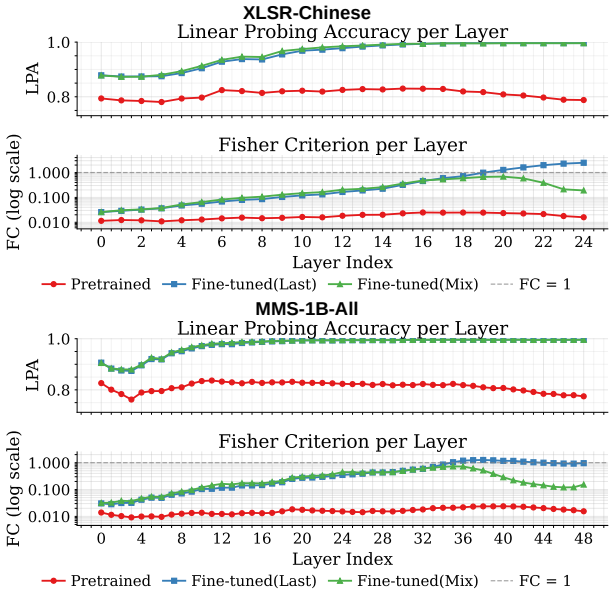


Figure 2: *Layer analysis results of XLSR-Chinese and MMS-1B-All under different training modes.*

First, we compare the LPA and FC of the XLSR-Chinese model in different layers, that is, directly extracting the embedding vectors of a single layer in the middle for logistic regression and Fisher's linear discriminant (Fig. 2). In terms of FC, the last-layer fine-tuned model separates tones better in the final layers, whereas the weighted-mix fine-tuned model achieves its best tone separation around layers 18 to 20; both outperform the pre-trained weights. In terms of LPA, the two fine-tuning modes show no notable difference, and both yield higher linear separability than the pre-trained weights at every layer. In both fine-tuned settings, the later layers show substantially higher FC compared to earlier layers, whereas in the pre-trained setting, this trend is only marginal.

MMS shows roughly the same layer-wise trends in LPA and FC. A notable difference is that, for the last-layer fine-tuned MMS model, FC rises to slightly above 1 at around layer 36 and stops increasing thereafter.

To analyze the effect of the projector, we take XLSR-Chinese as an example and apply the same analysis described above to the representation after the projector. The FC increases from 2.18 in the final layer to 10.99 after the projector, while the LPA increases only from 98.35% to 98.39% and the NCA from 98.30% to 98.39%, both close to ceiling. We attribute this to the bottleneck of the overall model lying in the CTC stage, so that the prototype built from the feature extractor's output vector and the one built from the vector after the projector show no noticeable difference. Nevertheless, the above information demonstrates that while the projector significantly reduces the information dimensionality, it also compresses the space of the same tone more tightly and separates different tone spaces.

6. Conclusions and Future Works

We introduce the STONE Taiwanese Hokkien tone classification model architecture, which utilizes a pre-trained backbone combined with CTC-guided prototype learning to obtain soft-label tone classification results that more closely resemble real-world tone classification scenarios. Furthermore, experiments show that the backbone already adapted to Mandarin (XLSR-Chinese) gives the best tone classification performance after task-specific fine-tuning, which significantly improves the feature extractor's ability to process tone information. An ablation further shows that boundary quality, not the classification head, dominates tone accuracy: the projector leaves the error rates essentially unchanged but reshapes the representation space so that tones become far better separated, which is what makes the soft labels meaningful.

The present study has four main limitations. The training labels and the OOD references are derived from sandhi rules rather than annotated surface tones, so TER measures agreement with the rules rather than with the tones actually produced; the soft-label scores are derived only from distances in the learned prototype space and have not been validated against native-speaker judgments; OOD TER is computed only on length-matched utterances and therefore describes a selected

subset rather than end-to-end performance; and all numbers come from a single run, so differences of a few tenths of a percentage point should not be read as significant.

In the future, this soft-label classifier will be used to calculate the tone soft label for each syllable of the speech generated by TTS. The tone accuracy of the speech will be calculated in conjunction with the score and margin indicators. The effectiveness of the soft label will be verified by calculating the Pearson/Spearman correlation coefficient with human mean opinion scores (MOS). A reasonable benchmark for the tone sandhi accuracy analysis of Taiwanese Hokkien will also be established.

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8. References

[1] Y. Bengio, A. Courville, and P. Vincent, "Representation learning: A review and new perspectives," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 35, no. 8, pp. 1798–1828, 2013.

[2] M. McAuliffe, M. Socolof, S. Mihuc, M. Wagner, and M. Sonderegger, "Montreal Forced Aligner: Trainable Text-Speech Alignment Using Kaldi," in *Proc. Interspeech 2017*, 2017, pp. 498–502.

[3] A. Graves, S. Fernández, F. Gomez, and J. Schmidhuber, "Connectionist temporal classification: labelling unsegmented sequence data with recurrent neural networks," in *Proceedings of the 23rd International Conference on Machine Learning*, ser. ICML '06. New York, NY, USA: Association for Computing Machinery, 2006, p. 369–376. [Online]. Available: https://doi.org/10.1145/1143844.1143891

[4] L. Kürzinger, D. Winkelbauer, L. Li, T. Watzel, and G. Rigoll, *CTC-Segmentation of Large Corpora for German End-to-End Speech Recognition*. Springer International Publishing, 2020, p. 267–278. [Online]. Available: http://dx.doi.org/10.1007/978-3-030-60276-5_27

[5] S. wen Yang, P.-H. Chi, Y.-S. Chuang, C.-I. J. Lai, K. Lakhotia, Y. Y. Lin, A. T. Liu, J. Shi, X. Chang, G.-T. Lin, T.-H. Huang, W.-C. Tseng, K. tik Lee, D.-R. Liu, Z. Huang, S. Dong, S.-W. Li, S. Watanabe, A. Mohamed, and H. yi Lee, "Superb: Speech processing universal performance benchmark," 2021. [Online]. Available: https://arxiv.org/abs/2105.01051

[6] P. Khosla, P. Teterwak, C. Wang, A. Sarna, Y. Tian, P. Isola, A. Maschinot, C. Liu, and D. Krishnan, "Supervised contrastive learning," 2021. [Online]. Available: https://arxiv.org/abs/2004.11362

[7] K. Q. Weinberger and L. K. Saul, "Distance metric learning for large margin nearest neighbor classification," *J. Mach. Learn. Res.*, vol. 10, p. 207–244, Jun. 2009.

[8] F. Schroff, D. Kalenichenko, and J. Philbin, "Facenet: A unified embedding for face recognition and clustering," in *2015 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, June 2015, p. 815–823. [Online]. Available: http://dx.doi.org/10.1109/CVPR.2015.7298682

[9] J. Snell, K. Swersky, and R. S. Zemel, "Prototypical networks for few-shot learning," 2017. [Online]. Available: https://arxiv.org/abs/1703.05175

[10] W. S.-Y. Wang and C. Sun, Eds., *The Oxford Handbook of Chinese Linguistics*. Oxford University Press, 04 2015, pp. 34–44. [Online]. Available: https://doi.org/10.1093/oxfordhb/9780199856336.001.0001

[11] J. Norman, "The mǐn dialects in historical perspective," *Journal of Chinese Linguistics Monograph Series*, no. 3, pp. 323–358, 1991. [Online]. Available: http://www.jstor.org/stable/23827042

[12] Ministry of Education, R.O.C., "Guidelines for Tâi-lô marking," Dictionary of Common Taiwanese Hokkien, n.d., accessed: 2026-05-23. [Online]. Available: https://sutian.moe.edu.tw/zh-hant/piantsip/tailo-phiautsu-suatbing/

[13] R. L. Cheng, "Tone sandhi in taiwanese," 1968. [Online]. Available: https://api.semanticscholar.org/CorpusID:143490793

[14] Z. Dong, "Compilation and comparison of the phonological systems of four dialects in taipei city, tainan city, lukang, and yilan," *Journal of Hsinchu Teachers College*, pp. 31–64, 1991.

[15] C. H. Wei, "Taiwanese minnan token-like words," https://huggingface.co/datasets/sarahwei/Taiwanese-Minnan-Sutiau, 2024.

[16] i3thuan5, "Suísiann dataset: Taiwanese speech corpus," 2026. [Online]. Available: https://suisiann-dataset.ithuan.tw/

[17] R. O. Duda, P. E. Hart, and D. G. Stork, *Pattern Classification*, 2nd ed. New York: John Wiley & Sons, 2000.

[18] G. Alain and Y. Bengio, "Understanding intermediate layers using linear classifier probes," 2018. [Online]. Available: https://arxiv.org/abs/1610.01644

[19] A. Pasad, J.-C. Chou, and K. Livescu, "Layer-wise analysis of a self-supervised speech representation model," 2022. [Online]. Available: https://arxiv.org/abs/2107.04734

[20] A. Baevski, H. Zhou, A. Mohamed, and M. Auli, "wav2vec 2.0: A framework for self-supervised learning of speech representations," 2020. [Online]. Available: https://arxiv.org/abs/2006.11477

[21] A. Conneau, A. Baevski, R. Collobert, A. Mohamed, and M. Auli, "Unsupervised cross-lingual representation learning for speech recognition," 2020. [Online]. Available: https://arxiv.org/abs/2006.13979

[22] V. Pratap, A. Tjandra, B. Shi, P. Tomasello, A. Babu, S. Kundu, A. Elkahky, Z. Ni, A. Vyas, M. Fazel-Zarandi, A. Baevski, Y. Adi, X. Zhang, W.-N. Hsu, A. Conneau, and M. Auli, "Scaling speech technology to 1,000+ languages," 2023. [Online]. Available: https://arxiv.org/abs/2305.13516

[23] J. Grosman, "Fine-tuned XLSR-53 large model for speech recognition in Chinese," https://huggingface.co/jonatasgrosman/wav2vec2-large-xlsr-53-chinese-zh-cn, 2021.

[24] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," 2019. [Online]. Available: https://arxiv.org/abs/1711.05101

[25] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2017. [Online]. Available: https://arxiv.org/abs/1412.6980

[26] Y.-H. Chou, K. Chang, M.-J. Wu, W. Ou, A. W.-H. Bi, C. Yang, B. Y. Chen, R.-W. Pai, P.-Y. Yeh, J.-P. Chiang, I.-T. Phoann, W. Chang, C. Cui, N. Chen, and J. Shi, "Evaluating self-supervised speech models on a taiwanese hokkien corpus," 2023. [Online]. Available: https://arxiv.org/abs/2312.06668